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H2 Mathematics (9758)
2026

2026 H2 Math Vectors I Homework 2: Solutions

Scalar product: properties, angles, perpendicularity, projections, ratios. Official school mark schemes where available; otherwise worked solutions. Q1 True/False answers from the Basics Worksheet. No vector-product methods are used. Total: 42 marks (Q1 unmarked).

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. $a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$	**T**
(b)	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$. $\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=\lvert\mathbf{a}\rvert^2$	**T**
(c)	$\mathbf{i}\cdot\mathbf{j}=0$. $1\cdot0+0\cdot1+0\cdot0=0$	**T**
(d)	$\mathbf{a}\cdot\mathbf{b}$ is a vector. $\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar)	**F**
(f)	If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$. $\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$	**T**
(g)	$(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$. $(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\,(\mathbf{a}\cdot\mathbf{b})$	**T**
(h)	$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector. $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$	**T**
(i)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$. Definition of scalar product	**T**

	Statement	**T/F**
(j)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}\cdot$ $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar)	**F**
(k)	If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$	**F**
(l)	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\cdot\mathbf{j}\rvert=0\neq1=\lvert\mathbf{i}\rvert\lvert\mathbf{j}\rvert$	**F**
(m)	$\mathbf{a}\cdot\mathbf{b}\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$	**T**
(n)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is acute or zero ($0^\circ\leq\theta<90^\circ$). $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta>0\ \therefore\ \cos\theta>0,\ 0^\circ\leq\theta<90^\circ$	**T**
(o)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is obtuse or 180° ($90^\circ<\theta\leq180^\circ$). $\cos\theta<0\ \therefore\ 90^\circ<\theta\leq180^\circ$	**T**
(q)	$\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore\ \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless	**F**
(r)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}:\ 4\neq2$	**F**
(s)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel. $\cos\theta=1\ \therefore\ \theta=0^\circ$	**T**
(t)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$. $\cos(360^\circ-\theta)=\cos\theta$	**T**

	Statement	**T/F**
(u)	Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at B	**T**
(v)	The length of projection of **a** onto **b** is equal to the length of projection of **b** onto **a**. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}:\ \lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert=1\neq 2=\lvert\mathbf{b}\cdot\hat{\mathbf{a}}\rvert$	**F**
(w)	If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2=0$	**T**
(x)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}:\ \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$	**F**
(y)	Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$	**F**
(z)	Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$. $\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$	**F**

2 [EJC 2024 Promo Q5]

$\mathbf{a}$ and $\mathbf{b}$ are vectors such that $|\mathbf{a}| = \sqrt{2}$ and $|\mathbf{b}| = \sqrt{6}$. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{5\pi}{6}$.

(a) Find the exact length of projection of $\mathbf{a}$ onto $\mathbf{b}$. [2]

(b) By considering $(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})$, find the exact value of $|3\mathbf{a}+2\mathbf{b}|$. [4]

Solution		Comments
(a) **[2]**	$\mathbf{a}\cdot\mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{5\pi}{6}$ $= \sqrt{2}\cdot\sqrt{6}\cdot\left(-\frac{\sqrt{3}}{2}\right) = -3$ Length of projection of $\mathbf{a}$ onto $\mathbf{b} = \frac{\lvert\mathbf{a}\cdot\mathbf{b}\rvert}{\lvert\mathbf{b}\rvert}$ $= \frac{\lvert-3\rvert}{\sqrt{6}} = \frac{3}{\sqrt{6}} = \frac{\sqrt{6}}{2}$.	
(b) **[4]**	$(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b}) = 9\lvert\mathbf{a}\rvert^2+12\mathbf{a}\cdot\mathbf{b}+4\lvert\mathbf{b}\rvert^2$ $= 9(2)+12(-3)+4(6)$ $= 18-36+24 = 6$ $\lvert3\mathbf{a}+2\mathbf{b}\rvert = \sqrt{6}$.	

3 [SAJC 2024 Promo Q9(a)]

Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

(i) By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. [3]

(ii) Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

(iii) The points P, Q and R have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of λ. [3]

Solution		Comments
(i) **[3]**	$(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})=\|2\mathbf{a}+\mathbf{b}\|^2$ $4\|\mathbf{a}\|^2+4\mathbf{a}\cdot\mathbf{b}+\|\mathbf{b}\|^2=(2\sqrt{74})^2=296$ $4(81)+4\mathbf{a}\cdot\mathbf{b}+1=296$ $325+4\mathbf{a}\cdot\mathbf{b}=296$ $4\mathbf{a}\cdot\mathbf{b}=-29$ $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$ (shown).	
(ii) **[1]**	Since $\|\mathbf{b}\|=1$, $\|\mathbf{a}\cdot\mathbf{b}\|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.	
(iii) **[3]**	$\overrightarrow{PQ}=(6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b})=-\mathbf{a}+10\mathbf{b}$ $\overrightarrow{PR}=(9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b})=2\mathbf{a}+(\lambda+5)\mathbf{b}$. Since P, Q, R are collinear and $\mathbf{a}$, $\mathbf{b}$ are non-parallel, $\overrightarrow{PR}=k\overrightarrow{PQ}$ for some scalar k: $2\mathbf{a}+(\lambda+5)\mathbf{b}=k(-\mathbf{a}+10\mathbf{b})$ $2=-k,\quad \lambda+5=10k$ $k=-2,\quad \lambda+5=-20$ $\lambda=-25.$	

4 [CJC 2022 Promo Q9(i),(ii)]

Relative to the origin O, the position vectors of points A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{b}$ is a unit vector. It is given that $\mathbf{b}$ and $\mathbf{b}-\mathbf{a}$ are perpendicular and C lies on AB such that $AC:CB=3:1$.

(i) Show that $|\mathbf{a}|=\sec\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. [3]

By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$,

(ii) find the value of $|\mathbf{c}\cdot\mathbf{b}|$ and state the geometrical interpretation of $|\mathbf{c}\cdot\mathbf{b}|$. [4]

Solution		**Comments**
(i) **[3]**	$\mathbf{b}\cdot(\mathbf{b}-\mathbf{a})=0$ $\lvert\mathbf{b}\rvert^2-\mathbf{a}\cdot\mathbf{b}=0$ $\mathbf{a}\cdot\mathbf{b}=1$ (since $\mathbf{b}$ is a unit vector) $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta=1$ $\lvert\mathbf{a}\rvert\cos\theta=1$ $\lvert\mathbf{a}\rvert=\sec\theta$ (shown).	
(ii) **[4]**	By the ratio theorem, $\mathbf{c}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$. $\mathbf{c}\cdot\mathbf{b}=\left(\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}\right)\cdot\mathbf{b}=\frac{1}{4}\mathbf{a}\cdot\mathbf{b}+\frac{3}{4}\lvert\mathbf{b}\rvert^2=\frac{1}{4}(1)+\frac{3}{4}=1$ $\lvert\mathbf{c}\cdot\mathbf{b}\rvert=1$. $\lvert\mathbf{c}\cdot\mathbf{b}\rvert$ is the length of the projection of $\mathbf{c}$ onto $\mathbf{b}$.	

5 [SAJC 2025 Promo Q8(i)]

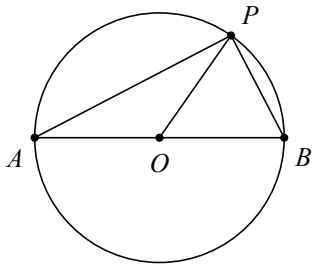
Relative to the origin O, two points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$, $|\mathbf{b}|=1$ and $|3\mathbf{a}-2\mathbf{b}|=\sqrt{37}$.

(i) By considering the scalar product $(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$, and give the geometrical meaning of $|\mathbf{a}\cdot\hat{\mathbf{b}}|$. [4]

Solution		**Comments**
(i) **[4]**	$(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})=\|3\mathbf{a}-2\mathbf{b}\|^2$ $9\|\mathbf{a}\|^2-12\mathbf{a}\cdot\mathbf{b}+4\|\mathbf{b}\|^2=37$ $9(4)-12\mathbf{a}\cdot\mathbf{b}+4(1)=37$ $40-12\mathbf{a}\cdot\mathbf{b}=37$ $\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$ (shown). $\|\mathbf{a}\cdot\hat{\mathbf{b}}\|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.	

6 [EJC 2025 Prelim P1 Q6]

There are three distinct points A, B and P. The point P lies on the circle with centre O and diameter AB. Relative to the origin O, the position vectors of points A and P are **a** and **p** respectively.



(a) By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of **a** and **p**, show that $\overrightarrow{AP}\cdot\overrightarrow{BP}=0$. [3]

The point C with position vector **c** lies on AB produced such that $AC:AB$ is $\lambda:1$.

(b) Find **c** in terms of λ and **a**. [2]

(c) It is given that PC is a tangent to the circle and that angle AOP is 120°. Using a suitable scalar product, find the value of λ. [4]

Solution		Comments
(a) **[3]**	$\overrightarrow{AP}=\mathbf{p}-\mathbf{a}, \qquad \overrightarrow{BP}=\mathbf{p}+\mathbf{a}$ (since $\overrightarrow{OB}=-\mathbf{a}$) $\overrightarrow{AP}\cdot\overrightarrow{BP}=(\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}+\mathbf{a})=\lvert\mathbf{p}\rvert^2-\lvert\mathbf{a}\rvert^2=0$ (since $\lvert\mathbf{a}\rvert=\lvert\mathbf{p}\rvert=$ radius).	
(b) **[2]**	Since $AC:AB=\lambda:1$ and $\overrightarrow{AB}=-2\mathbf{a}$, $\overrightarrow{AC}=\lambda\overrightarrow{AB}=-2\lambda\mathbf{a}$ $\mathbf{c}=\mathbf{a}+\overrightarrow{AC}=(1-2\lambda)\mathbf{a}.$	
(c) **[4]**	Since PC is a tangent, $\overrightarrow{OP}\perp\overrightarrow{PC}$: $\mathbf{p}\cdot((1-2\lambda)\mathbf{a}-\mathbf{p})=0$ $(1-2\lambda)\,\mathbf{p}\cdot\mathbf{a}-\lvert\mathbf{p}\rvert^2=0$ $(1-2\lambda)\,\lvert\mathbf{a}\rvert\lvert\mathbf{p}\rvert\cos 120^\circ-\lvert\mathbf{a}\rvert^2=0$ $(1-2\lambda)\,\lvert\mathbf{a}\rvert^2\left(-\frac{1}{2}\right)-\lvert\mathbf{a}\rvert^2=0$ $-\frac{1}{2}(1-2\lambda)=1$ $1-2\lambda=-2$ $\lambda=\frac{3}{2}.$	

7 [NJC 2025 Prelim P2 Q3]

With respect to an origin O, points P and Q have variable position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively, given by

$$\mathbf{p} = (\cos t)\,\mathbf{i} + (\sin t)\,\mathbf{j} - \mathbf{k}, \qquad \mathbf{q} = (\cos 2t)\,\mathbf{i} - (\sin 2t)\,\mathbf{j} + \alpha\mathbf{k},$$

where α is a real parameter and $0 < t < \pi$.

(i) Find the exact values of t and α if O, P and Q are collinear. [5]

It is given that $\alpha = 0.5$.

(ii) Show that $\mathbf{p} \cdot \mathbf{q} = \cos 3t - 0.5$. Find the greatest length of projection of $\mathbf{q}$ onto $\mathbf{p}$. [4]

Solution		Comments
(i) **[5]**	Since O, P, Q are collinear, $\mathbf{q} = k\mathbf{p}$ for some scalar k: $\begin{pmatrix} \cos 2t \\ -\sin 2t \\ \alpha \end{pmatrix} = k \begin{pmatrix} \cos t \\ \sin t \\ -1 \end{pmatrix}$. From the **j**-component: $-\sin 2t = k \sin t$ $-2\cos t \sin t = k \sin t$ $\sin t(-2\cos t - k) = 0$ $\sin t = 0$ (reject, since $0 < t < \pi$) or $\cos t = -\frac{k}{2}$. From the **i**-component: $\cos 2t = k \cos t$ $2\cos^2 t - 1 - k\cos t = 0$. Substitute $\cos t = -\frac{k}{2}$: $2\left(-\frac{k}{2}\right)^2 - k\left(-\frac{k}{2}\right) - 1 = 0$ $\frac{k^2}{2} + \frac{k^2}{2} - 1 = 0$ $k^2 = 1$ $k = \pm 1$. From the **k**-component, $\alpha = -k$. $k = 1 \Rightarrow \cos t = -\frac{1}{2} \Rightarrow t = \frac{2\pi}{3}$ $(0 < t < \pi)$, $\alpha = -1$ $k = -1 \Rightarrow \cos t = \frac{1}{2} \Rightarrow t = \frac{\pi}{3}$, $\alpha = 1$. Hence $t = \frac{\pi}{3}$, $\alpha = 1$ or $t = \frac{2\pi}{3}$, $\alpha = -1$.	

(ii) **[4]**	$\mathbf{p}\cdot\mathbf{q} = \cos t\cos 2t - \sin t\sin 2t - 0.5$ $= \cos(2t+t) - 0.5$ $= \cos 3t - 0.5$ (shown). $\lvert\mathbf{p}\rvert = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}$ Length of projection of $\mathbf{q}$ onto $\mathbf{p} = \dfrac{\lvert\mathbf{p}\cdot\mathbf{q}\rvert}{\lvert\mathbf{p}\rvert}$ $= \dfrac{\lvert\cos 3t - 0.5\rvert}{\sqrt{2}}$. The greatest value of $\lvert\cos 3t - 0.5\rvert$ on $0 < t < \pi$ is 1.5 (when $\cos 3t = -1$), so greatest length $= \dfrac{1.5}{\sqrt{2}} = \dfrac{3}{2\sqrt{2}}$ $(= 1.06$ (3 s.f.)).	

END OF SOLUTIONS